



Parametric Study of the Hypersonic Near-Field and Sonic Boom from Waveriders using a Fully-Parabolized Approach

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A parametric study is performed to understand the relationship between volume displacement, lift, near-field signature, and sonic boom overpressure for variable wedge angle power-law waveriders. The width of a parametric waverider is varied for freestream Mach numbers from 5 to 7. Both near-field and sonic boom predictions are made with a fully parabolized approach. The Upwind Parabolized Navier-Stokes solver is used to spatially march the hypersonic flow-field in the streamwise direction. The waveform parameter method is used to propagate the hypersonic near-field to the ground from a fixed altitude of 15.85 km. We find the magnitude of the SPL varies with frequency as $-19.7 \log f$. There is a positive quasi-linear relationship between near-field and sonic boom overpressures with volume displacement. For $M_\infty = 7$, a 150% volume increase yields 92.5% and 60.9% rises in near-field and sonic boom overpressures, respectively. The effect of losses due to thermo-viscous effects and atmospheric absorption are quantified. We show that for a waverider of volume $V = 4970 \text{ cm}^3$ at $M_\infty = 7$, these losses, predicted by PCBoom using modules PCBurg and enhanced Burgers' decrease maximum overpressure by 41.6% and 39.5% relative to WPM, respectively.

Nomenclature

Symbols	Description	Non-Dimensional Numbers
A	Vehicle width factor	c_l Lift coefficient
E, F, G	Streamwise flux vectors	c_d Drag coefficient
L	Body length	M Mach number
m	Slope of waveform segment	
p	Static pressure	Abbreviations
r	Radial coordinate	PNS Parabolized Navier-Stokes
r_s	Source cylinder radius	UPS Upwind Parabolized Navier-Stokes Solver
s	Ray path length	WPM Waveform Parameter Method
$\hat{p}$	Fourier transform of scaled acoustic pressure	
Greek Symbols		Superscripts
α_t	Doppler shifted absorption coefficient	p Elliptic regime
$\tilde{\beta}$	Coefficient of nonlinearity	$*$ Parabolic-hyperbolic regime
γ	Ratio of specific heats	
ξ, η, ζ	Transformed coordinates	Subscripts
λ	Time duration of waveform segment	∞ Ambient condition
ω	Frequency	
θ	Surface inclination angle	

I. Introduction

Hypersonic waveriders and interceptors are the subject of intense research, development, and deployment. Hypersonic flight-vehicles create unique signatures in the form of sonic booms. The aerodynamic near-field and sonic boom of a hypersonic slender body is unique relative to blunt bodies and slender supersonic bodies. In particular, the gas flowing over a hypersonic wave rider may partially or fully ionize, which requires additional or modified closure equations. Furthermore, unlike low-speed supersonic near-field calculations, viscous effects must be incorporated for accurate near-field and sonic boom predictions. There are very few large-scale studies available to the public focusing on design

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spaces of hypersonic flight-vehicles and their effect on sonic boom. In this paper, we present predictions and methods to create a numerical database for hypersonic leading geometries. We focus on the leading geometry because it creates the dominant leading shock wave overpressure in both the near-field and ground waveforms.

King and Miller [1] and King et al. [2] present a fully-parabolized approach for predicting hypersonic sonic boom from slender bodies. In this approach, a parabolized Navier-Stokes solver with corresponding gas models spatially march the flow-field from the vehicle surface to a source cylinder. A sonic boom solver, the waveform parameter method (WPM), then propagates the near-field signature to the ground. This propagation solver is advantageous because it is extremely fast and completely integrated within the parabolized solver. The combination of both approaches yields a complete spatial marching method from flight-vehicle surface to ground observer.

Particular terms in the equations of motion have negligible effect on the near-field and sonic boom relative to others. These terms can be eliminated without sacrificing accuracy, which yields the advantage of reducing computational expense. To minimize computational expense without sacrificing accuracy, King et al. [2] showed that some viscous terms have greater influence on near-field and sonic boom overpressures as M_∞ increases. Narain et al. [3] used a parabolized approach to study complex maneuvering reentry vehicle configurations and showed good agreement with flight test measurements. Real gas and non-equilibrium models are also included in the present study that are necessary for hypersonic flow-fields, although these terms are found to be less significant than viscous terms [1, 2].

Today, fully nonlinear CFD methods dominate the field of sonic boom prediction. For example, one leading CFD solver is NASA's Fully-Unstructured Navier-Stokes (FUN3D) [4], which is more accurate than the present approach in the near-field. FUN3D requires an initial three-dimensional computational domain and uses adjoint based grid adaptation to resolve shock waves. However, FUN3D on a per case basis uses more CPU time relative to the present parabolized approach. Furthermore, the approach of King and Miller [1] generates the computational grid during run-time based on a parameterization or initial flight-vehicle surface grid. King and Miller [1] propagate the surface grid in the direction of the radiating shock waves, following the approach of Housman et al. [5]. This is essential for the accurate prediction of the near-field pressure signature.

Traditional analytical approaches include the F-function of Whitham [6], impulse method of Tiegerman [7], and the newly developed approach of Miller [8]. Though these methods have the benefit of being closed-form equations for the near-field and sometimes ground waveform, they do not explicitly account for important physics like boundary layer growth without empirical corrections. While fully closed-form analytical solutions are the fastest and in the opinion of the authors' most elegant, they may not be suitable for large-scale parametric studies that include real gas and viscous effects. The fully-parabolized approach presented here represents an intermediate approach.

Note that we do not examine atmospheric variations in our study, which can have a large-effect on sonic boom propagation (see Pierce [9], Stout et al. [10], and Carr et al. [11]). Previous studies by King et al. [2], and others have examined the impact of atmospheric variations on hypersonic sonic boom propagation, where factors such as wind, pressure, temperature, and turbulence were included.

There are open questions regarding how the near-field and sonic boom are altered by lift, drag, atmospheric turbulence, shape of the leading edge, and other parameters. In this paper, we present a methodology for finding these trends quickly for one particular analytical waverider leading geometry and related aerodynamic coefficients. In particular, we are interested in lift coefficient, c_L , and flight-vehicle width, which is controlled by one parameter, A . We seek to find correlations via numerical predictions between waverider geometry and aerodynamic properties with sonic boom signature.

The paper is laid out as follows. First, we present the methodology, which consists of the near-field parabolized solver, the WPM sonic boom propagation solver, and PCBoom. We then present results that include aerodynamics of the near-field and ground waveform for variable wedge angle (VWA) power-law waveriders. Losses due to thermo-viscous effects and absorption in the atmosphere are quantified. We provide an analysis of the trends regarding the connection between waverider width, lift, near-field signature, and sonic boom signature. Finally, a summary and conclusion are presented.

II. Methodology

We predict hypersonic flow-fields via a fully-parabolized spatial marching solver. As previously noted, this solver propagates the radiating waves from the aerodynamic field to the near-field, and then to the ground via the WPM. Figure 1 presents the prediction approach. Here, the hypersonic flow-field (top), the near-field source cylinder pressure (middle), and the sonic boom (bottom) are shown. Flow moves from left to right. The source cylinder is located at radius r_s . The mathematical surface enclosing the flight-vehicle, where pressure is measured or predicted in the near-field, is

represented by the source cylinder. Solid and dashed lines depict strong leading and trailing shock waves, respectively. Numerous shock and expansion waves exist between the leading and trailing shock waves.

In the parabolized method, solutions progress downstream and terminate upon intersecting the source cylinder at location, r_s , and azimuthal angle, ϕ . Therefore, we do not prescribe a downstream boundary condition. We ensure the near-field source cylinder is located in the range defined by Loubeau and Coulouvrat [12] through previously performed numerical studies. L is the vehicle length. The near-field represents the region as close to the flight-vehicle as possible, but outside the region of induced vorticity. The parabolized approach marches downstream from a prescribed free-stream boundary condition. The free-stream boundary condition is defined by the global Mach number of the flight-vehicle, M_∞ , the ambient pressure at the flight-vehicle altitude, p_∞ , and the ambient temperature, T_∞ . Predictions are normalized in the near-field by p_∞ . It is common for hypersonic flight-vehicles, even at low altitudes, to generate intense over-pressure as N-waves at the ground, as depicted at the bottom of Fig. 1.

The parabolized solver requires the surface to be parameterized or be generated from a surface grid. At minimum for the power-law waverider geometry, the flight-vehicle geometry and body length, L , must be defined. The minimum computational grid spreading angle is constrained by the local angle of the leading shock wave. The solver generates the computational domain algebraically as the solution is marched downstream. The flow-field is integrated in space by solving the PNS equations, which are closed using equilibrium, finite-rate non-equilibrium, or frozen air chemistry models [13, 14].

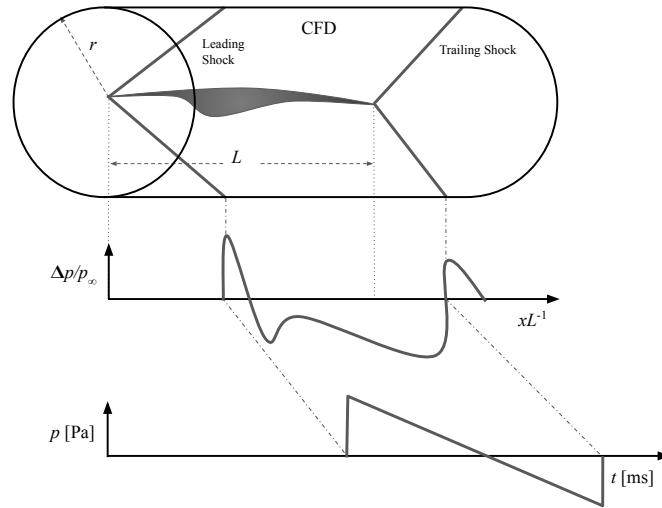


Fig. 1 Hypersonic flow-field, source pressure cylinder, and sonic boom at the ground [2].

A. Upwind Parabolized Navier-Stokes Solver

We present the parabolized Navier-Stokes (PNS) equations that model the hypersonic near-field and are solved numerically via a finite volume, spatial marching approach. The PNS equations are derived from the Navier-Stokes equations by assuming particular viscous derivatives to be negligible [15]. The PNS equations, in flux vector form, are

$$\frac{\partial \mathbf{E}}{\partial \xi} + \frac{\partial \mathbf{F}}{\partial \eta} + \frac{\partial \mathbf{G}}{\partial \zeta} = 0, \quad (1)$$

where $\mathbf{E}$, $\mathbf{F}$, and $\mathbf{G}$ are the flux vectors in the ξ , η , and ζ directions, respectively.

The PNS equations are parabolic-hyperbolic in the ξ -direction (streamwise direction) if the local flow-field is supersonic. In subsonic regions (elliptic) of the flow-field, departure behavior is restrained by employing the technique of Vigneron et al. [16]. An example of potential local elliptic regions of the flow-field is inside a boundary layer upstream of a corner. Vigneron's technique separates the streamwise flux vector into two terms, where $\mathbf{E}^P$ is the elliptic portion and $\mathbf{E}^*$ is the parabolic-hyperbolic portion. This is represented as $\mathbf{E} = \mathbf{E}^* + \mathbf{E}^P$. We then discretize the split streamwise flux vector via forward differencing. The streamwise flux gradient is then expressed as

$$\left(\frac{\partial \mathbf{E}}{\partial \xi}\right)_{i+1} = \frac{1}{\Delta \xi} [\mathbf{E}_{i+1}^* - \mathbf{E}_i^* + \mathbf{E}^P(d\mathbf{S}_{i+1}, \mathbf{U}_{i+2}) - \mathbf{E}^P(d\mathbf{S}_i, \mathbf{U}_{i+1})]. \quad (2)$$

Streamwise and cross-stream flux gradients are then substituted into the PNS equations (see Eqn. 1) and the iterative PNS (IPNS) method of Miller et al. [15] is implemented in regions with upstream influence to improve predictions. The discretized form of the equations used is

$$\begin{aligned} & \left\{ \frac{1}{\Delta \xi} [A^*(d\mathbf{S}_{i+1}, \mathbf{U}_i) - A_i^P]^{k+1} + \frac{\partial}{\partial \eta} \left[\frac{\partial \mathbf{F}(d\mathbf{S}_{i+1}, \mathbf{U}_i)}{\partial \mathbf{U}} \right]^{k+1} + \frac{\partial}{\partial \zeta} \left[\frac{\partial \mathbf{G}(d\mathbf{S}_{i+1}, \mathbf{U}_i)}{\partial \mathbf{U}} \right]^{k+1} \right\} \Delta \mathbf{U}_i^{k+1} \\ &= -\frac{1}{\Delta \xi} \{ [A(d\mathbf{S}_{i+1}, \mathbf{U}_i) - A_i]^{k+1} \mathbf{U}_i^{k+1} - \mathbf{E}^P(d\mathbf{S}_{i+1}, \mathbf{U}_{i+2})^k + \mathbf{E}^P(d\mathbf{S}_{i+1}, \mathbf{U}_i)^{k+1} \} \\ & \quad - \left[\frac{\partial \mathbf{F}(d\mathbf{S}_{i+1}, \mathbf{U}_i)}{\partial \eta} \right]^{k+1} - \left[\frac{\partial \mathbf{G}(d\mathbf{S}_{i+1}, \mathbf{U}_i)}{\partial \zeta} \right]^{k+1}. \end{aligned} \quad (3)$$

The flow-field solver, Upwind Parabolized Navier-Stokes Solver (UPS), was developed primarily for flight-vehicle aerodynamic prediction, and it is based on the work of Lawrence et al. [17]. UPS is capable of simulating ideal gas, real gas, and non-equilibrium reacting flow, as shown by Rubin and Tannehill [18].

The real gas model, which was developed by Tannehill et al. [14], is utilized by the UPS system to simulate chemically reacting flows. In the case of non-equilibrium flow-fields, the species continuity equation, as proposed by Tannehill et al. [13], is numerically evaluated. By doing so we account for O_2 , N_2 , O , N , NO , NO^+ , and e^- varying species. The details of this equation and variables are discussed in Tannehill et al. [13] and Tannehill et al. [14]. The accuracy of near-field predictions using the UPS system has been quantified by King and Miller [1].

B. Waveform Parameter Method

We predict sonic boom via WPM, which was originally proposed by Thomas [19]. Propagation paths are modeled via ray theory through the varying meanflow of the atmosphere. WPM is based on three fundamental differential equations. The first differential equation models the evolution of the local slope of each waveform segment, m_i , with respect to waveform time, t , as

$$\frac{dm_i}{dt} = c_1 m_i^2 + c_2 m_i, \quad (4)$$

where i represents the segment. The increase in pressure due to shock waves is denoted as Δp . Change in pressure is

$$\frac{d\Delta p_i}{dt} = \frac{1}{2} c_1 \Delta p_i (m_i + m_{i-1}) + c_2 \Delta p_i. \quad (5)$$

Finally, the duration of each segment, denoted as λ_i , evolves according to

$$\frac{d\lambda_i}{dt} = -\frac{1}{2} c_1 (\Delta p_i + \Delta p_{i+1}) - c_1 m_i \lambda_i. \quad (6)$$

The auxiliary functions, c_1 and c_2 , are defined by Thomas [19]. Here, A represents the local ray-tube area at the waveform position along the ray path, c is the local ambient speed of sound, γ is the ratio of specific heats, and ρ is the local density. In addition to predicting time varying waveforms at the ground, sonic boom signatures are also presented as sound pressure level (SPL). SPL is normalized by a reference pressure of $20 \mu\text{Pa}$. WPM has previously been validated against PCBoom of Plotkin et al. [20] and propagation results from the Second AIAA/NASA Sonic Boom Prediction Workshop [21].

C. PCBoom

We compare predictions from UPS and WPM to NASA's PCBoom [22] for validation. We also use PCBoom to conduct a study to quantify loss effects. We use three approaches with PCBoom. The first is FOBoom, which is a lossless solver. Second, PCBurg, which accounts for some losses using the FOBoom solver. Third, we use the PCBoom enhanced Burgers' equation solver.

We first generate solutions with the FOBoom module within the PCBoom suite using CFD predictions obtained from UPS. FOBoom propagates the source waveform along a ray using thin shock wave theory. We then incorporate atmospheric effects with PCBurg, which creates a modified waveform from the Burgers' equation. PCBurg uses the Burgers' equation

$$\frac{\partial p}{\partial t} + \frac{\gamma + 1}{2\gamma} \frac{c_\infty}{p_\infty} p \frac{\partial p}{\partial \xi} = \text{IFT} [\alpha(\omega) \text{FT}[p]], \quad (7)$$

where α is frequency dependent atmospheric absorption, $\text{FT}[]$ denotes the Fourier transform operator, $\text{IFT}[]$ denotes the inverse Fourier transform operator, and ω is the angular frequency. To account for three-dimensional effects, the acoustic pressure is multiplied by a scaling factor at each step. Finally, we use the enhanced Burgers' solver to predict the ground signature [23]. The enhanced Burgers' solver within PCBoom includes nonlinear thermo-viscous effects in an improved formulation relative to PCBurg. The governing equation for the enhanced Burgers' solver is

$$\frac{\partial \hat{p}}{\partial s} = \frac{i\omega\tilde{\beta}}{4\pi} \int \hat{p}(\omega') \hat{p}(\omega - \omega') d\omega' - \alpha_t(\omega) \hat{p}(\omega), \quad (8)$$

where s is the ray path, $\hat{p}$ is the Fourier transform of the scaled acoustic pressure, α_t is the Doppler shifted and convected absorption coefficient, $\tilde{\beta}$ is the coefficient of nonlinearity, and ω is the angular frequency, [24]. All sonic boom solvers propagate the signal from flight-vehicle to observer along the ray paths.

III. Results

We examine a variable wedge angle (VWA) power-law waverider. Four equations define the VWA geometry and are $z_{LE} = Ax^n$, $z_{FS} = By^n$, $z_{LS} = A(y - x \tan \theta)/(\tan \beta - \tan \theta)$, and $B = A/\tan^n \beta$, where z_{LE} , z_{FS} , and z_{LS} represent the z -coordinates for the leading edge, the freestream surface, and the lower surface, respectively [25]. In this paper, we set $A = 0.66$, $n = 0.5$, $\theta = 7^\circ$, and $\beta = 7^\circ$. We set $\beta = \theta$ to enforce a flat lower surface. The after-body converges to a cylindrical cross-section at -2° from 0.5 to 1.2 m. In the original study of Starkey and Lewis [25], the forebody of the vehicle is 50 m long. We scale the geometry by a factor of 10^{-2} from the original parameters of Starkey and Lewis [25]. This scaling is chosen so that the scale of the leading geometry is on the same order of magnitude as contemporary waveriders. A schematic of the waverider geometry with the surface grid is shown in Fig. 2, and Fig. 2b shows the top-view, Fig. 2c shows the side view, and Fig. 2a shows the front view. Every 40th streamwise grid point is presented.

Grid independence studies were previously conducted by King et al. [2]. 800 radial grid points and 45 azimuthal grid points showed a relative difference of 0.57% in maximum overpressure compared to the finest case. This agreed with the results of King and Miller [1], who performed a grid independence study on an axisymmetric cone-cylinder-cone. UPS was also validated by King and Miller [1] with the theory of Taylor and Maccoll [26] and Taylor [27] for inviscid conical flows at zero angle-of-attack. Supersonic and hypersonic boundary layer validation of UPS with the results of Oz and Kursat [28] and Esquieu et al. [29], respectively, were performed by King and Miller [1]. WPM was validated by King et al. [2] with PCBoom, and cases from the Second AIAA/NASA Sonic Boom Prediction Workshop [21]. Maximum relative difference of sonic boom overpressure with PCBoom was within a maximum of 8.0% for all test cases.

To demonstrate the approach, we predict $M_\infty = 7.0$ flow over the VWA waverider for inviscid flow with an ideal gas and an inviscid non-equilibrium real gas. The atmospheric conditions are $p_\infty = 12.13$ kPa, $\rho_\infty = 0.1948$ kg m⁻³, and $T_\infty = 217.0$ K. Figure 3 shows $\Delta p p_\infty^{-1}$ contours. The waverider has a sharp leading edge. The freestream condition is denoted as light blue and maximum pressure ($\Delta p p_\infty^{-1} \approx 1.8$) along the vehicle under-track is denoted as light yellow. We note a shock wave that is attached to the leading edge.

The near-field source cylinder is extracted at $rL^{-1} = 0.1667$. Ideally, a source cylinder radius should be greater than $rL^{-1} = 0.20$; however, this is still within the range studied previously by Loubeau and Coulouvrat [12]. The near-field begins where the induced vorticity of the flight-vehicle is less than 1% of its maximum.

Figure 4a shows this source cylinder pressure, $\Delta p p_\infty^{-1}$, versus waverider length normalized x -position, xL^{-1} . The freestream condition occurs as $\Delta p p_\infty^{-1}$ is zero before encountering the leading shock wave. For the case of inviscid and non-equilibrium flow with a Mach number of $M_\infty = 5.0$, the predicted normalized near-field overpressure in Fig. 4a is

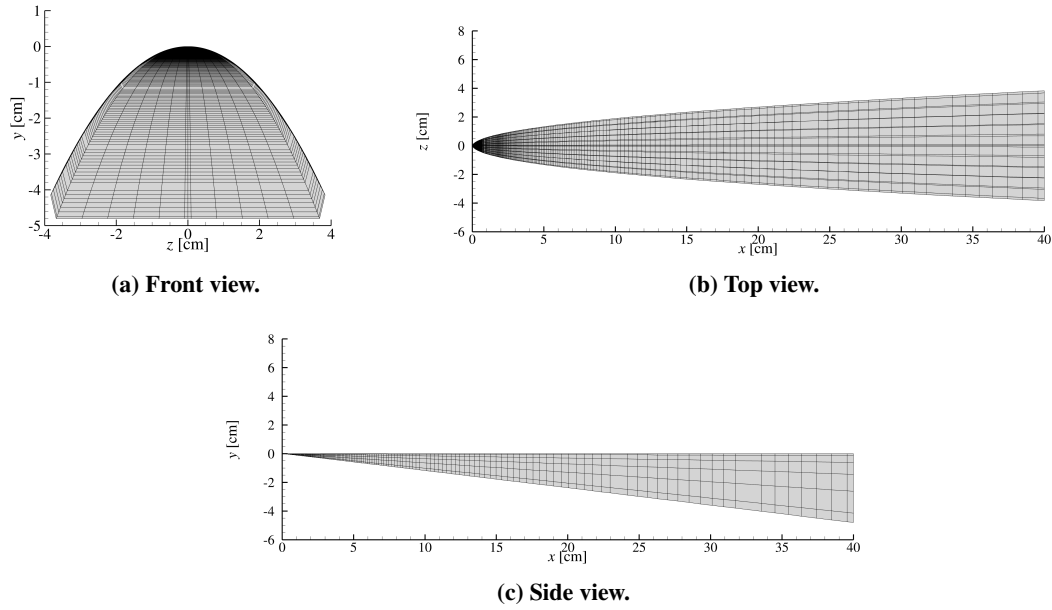


Fig. 2 Power-law waverider fore-body surface grid.

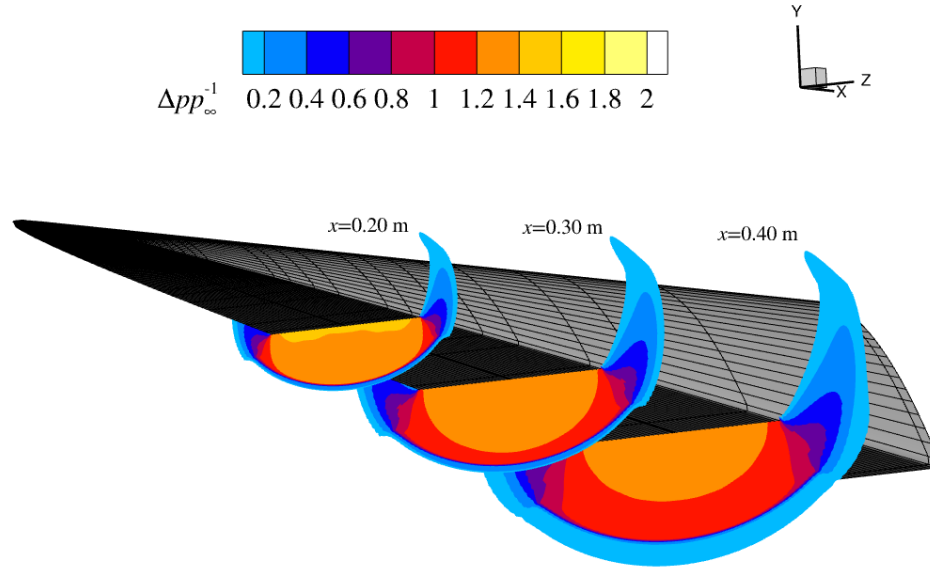


Fig. 3 Contours of $\Delta p p_{\infty}^{-1}$ for the power-law waverider [2].

found to be 0.452 and 0.439, respectively. Under the assumption of an ideal gas, there is a relative increase of 2.96% between the overpressures. At $M_{\infty} = 7.0$, the leading shock wave, which occurs at $xL^{-1} \approx 0.78$ on the source cylinder, creates a p_{∞} -normalized overpressure in Fig. 4a of 0.811 and 0.757 for inviscid and non-equilibrium flow, respectively. Expansion waves propagating from the turning corner between forebody and afterbody decrease the near-field pressure to $\Delta p p_{\infty}^{-1} = -0.309$ at $xL^{-1} = 1.62$ for inviscid flow and $\Delta p p_{\infty}^{-1} = -0.316$ at $xL^{-1} = 1.59$ for non-equilibrium flow.

We propagate the near-field pressure to the ground from an altitude of 15.85 km via WPM. We assume a standard, quiescent atmosphere with a ground reflection factor of 1.9. Figure 4b shows the sonic boom signature at the ground for $M_{\infty} = 5$. The sonic boom signature takes the form of an N-wave. For an ideal gas and non-equilibrium flow, sonic

boom predicted overpressure is 4.37 and 4.39 Pa, respectively. This is a relative difference of 0.42% [2]. Shocks and expansions coalesce as the waves propagate through the atmosphere. We now propagate the near-field at $M_\infty = 7.0$ to the ground from the same altitude for an ideal gas and non-equilibrium flow. Here, the sonic boom overpressure in Fig. 4b is 4.83 and 4.81 Pa for an ideal gas and non-equilibrium flow, respectively. This represents a relative difference of 0.47% [2]. The greatest difference between the two sonic boom waveforms is their duration. For an ideal gas at $M_\infty = 5$ and $M_\infty = 7$, the duration of the sonic boom is 9.91 and 10.77 ms, respectively. For non-equilibrium flow at $M_\infty = 5$ and $M_\infty = 7$, the duration is 10.31 and 10.95 ms, respectively. On average, this represents a 2.8% difference [2].

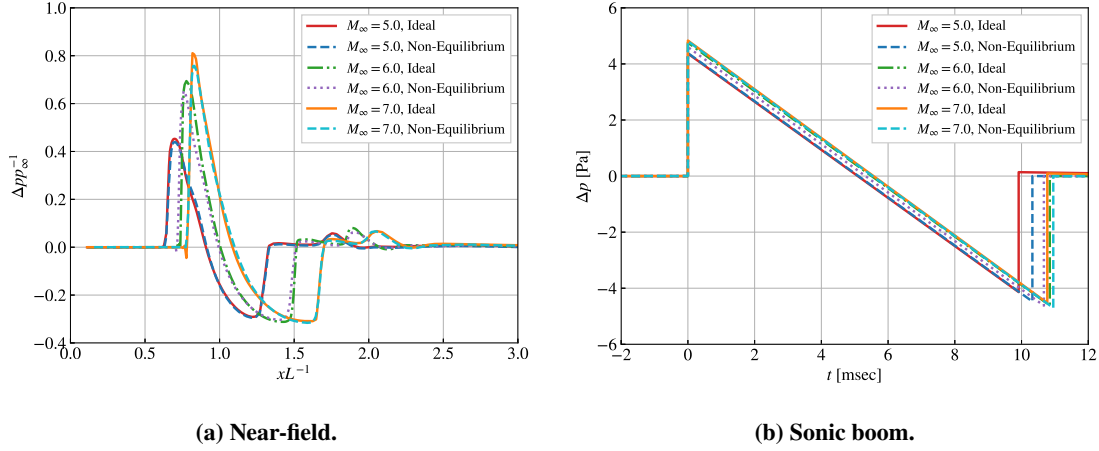


Fig. 4 Near-field and sonic boom waveforms for the power-law waverider when $A = 0.66$, $n = 0.5$, and $\theta = 7$ deg.

The relationship between sonic boom duration and ground spectra is examined. The ground pressure waveform serves as the basis for the Sound Pressure Level (SPL) calculation, and we focus on base cases with parameters $A = 0.66$, $n = 0.5$, and $\theta = 7$ deg. Figure 5 shows the SPL per unit Hz at the ground observer, which shows local maxima for each sonic boom signature. For $M_\infty = 5$, the left side of Fig. 5 shows the SPL, while the right side shows $M_\infty = 7$. Ideal gas predictions are represented by black lines with downward-facing triangles, contrasted with blue lines and upward-facing triangles for non-equilibrium flow. Notably, at $M_\infty = 5$, ideal gas flow yields a peak SPL of 94.63 dB at 100.8 Hz, and non-equilibrium flow results in 93.81 dB at 97.7 Hz. For $M_\infty = 7.0$, inviscid flow reaches a peak SPL of 92.42 dB at 91.5 Hz, whereas non-equilibrium flow reaches 94.82 dB at 69.3 Hz. Non-equilibrium flow at $M_\infty = 7.0$ extends sonic boom duration by 3.9%, simultaneously lowering peak SPL frequency by 24.3% [2]. The SPL spectral comparison at $M_\infty = 5$ demonstrates a closer alignment between ideal and non-equilibrium flows than that observed at $M_\infty = 7$. This discrepancy indicates a deviation from ideal gas predictions as hypersonic M_∞ increases, as observed in sonic boom duration predictions (Figs. 4b and 4b). Non-equilibrium conditions prolong the sonic boom duration, leading to a reduced fundamental frequency and a decrease in peak SPL frequency [2].

There is a clear power-law relationship in which the sonic boom waveform's pressure spectra collapses. We observe that for the three ambient Mach numbers studied, SPL goes as $-19.7 \log f$. Thus, we also show that SPL spectra of the sonic boom waveform is also independent of M_∞ within the range studied.

A. Propagation Through a Realistic Atmosphere with Losses

Thermo-viscous effects on sonic boom propagation are studied. We compare WPM to PCBoom's lossless solver FOBoom, PCBurg, and enhanced Burgers' solver. We select three cases: $M_\infty = 5$ and $V = 828 \text{ cm}^3$, $M_\infty = 7$ and $V = 4970 \text{ cm}^3$, and $M_\infty = 6$ and $V = 2982 \text{ cm}^3$. Figures 6a through 6c show the sonic boom waveform prediction at the ground for these cases. The ground signatures are aligned at their respective rise times, when $\Delta p = 0.63 \Delta p_{\text{MAX}}$. Denoted by solid black lines are the predictions made by WPM, the dashed red lines represent FOBoom, the dashed-dotted blue lines represent PCBurg, and the dotted green lines represent the enhanced Burgers' solver. For the most extreme case in Fig. 6c, predicted maximum overpressures are 7.03 Pa for FOBoom, 4.04 Pa for PCBurg, and 4.19 Pa for the enhanced Burgers' solver. The percent decreases in maximum overpressure from the WPM to the PCBurg and the enhanced Burgers' solver were 41.6% and 39.5%, respectively, which demonstrates the thermo-viscous and atmospheric absorption effects on ground signature. Figure 6d shows the percent decrease in maximum overpressure with PCBurg and

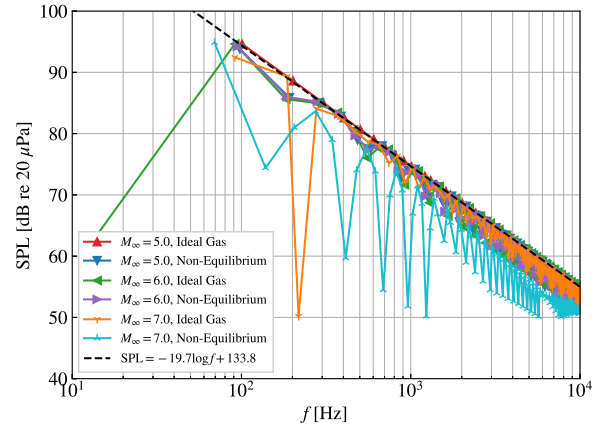


Fig. 5 Sound pressure level spectra of the sonic boom waveform at the ground.

the enhanced Burgers' solver relative to WPM. WPM and FOBoom will always predict a larger maximum overpressure due to it being a lossless solver. Overall, in terms of overpressure and duration, PCBurg tends to vary the greatest from the lossless predictions of WPM and FOBoom across the cases presented.

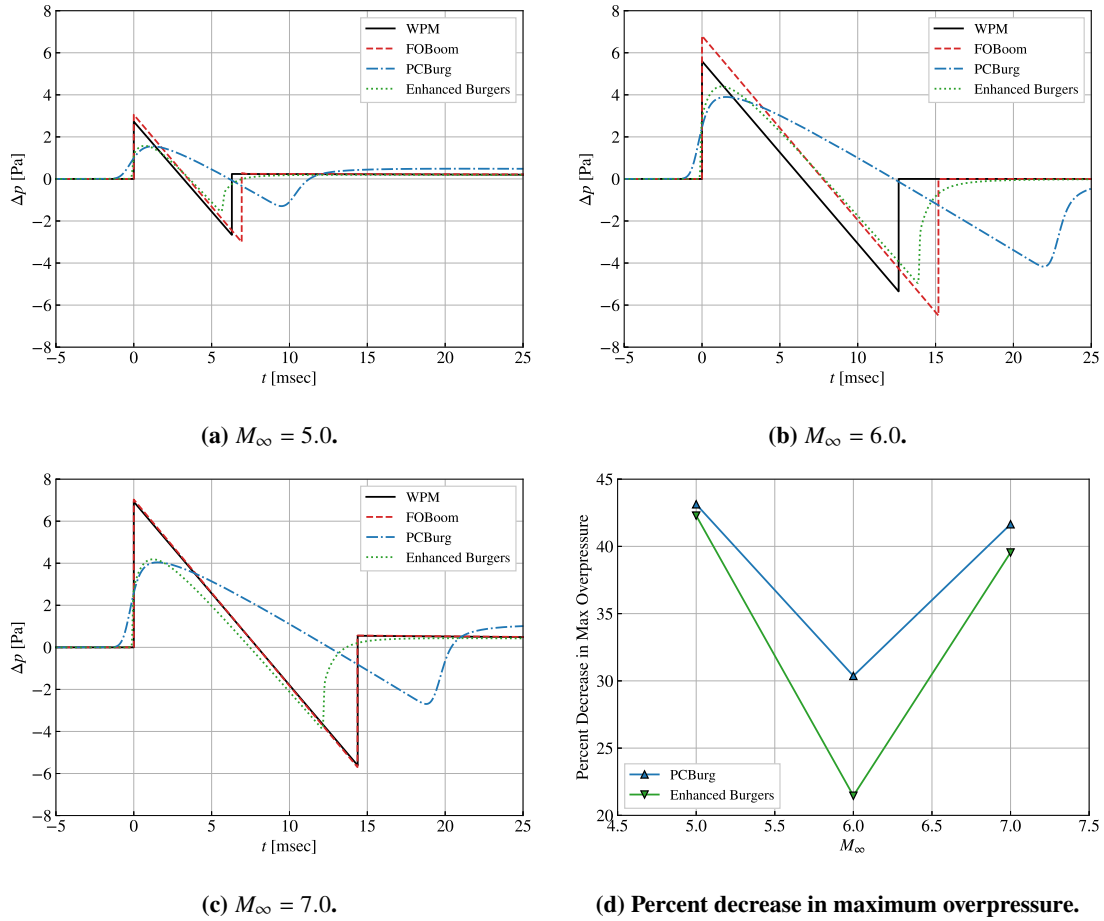


Fig. 6 Sonic boom waveforms predicted using WPM, FOBoom, PCBurg, and Enhanced Burgers'.

B. Effect of Vehicle Design Parameters on Near-Field and Sonic Boom Waveforms

We use the base case of the waverider designed using the parameters given by Starkey and Lewis [25] and vary M_∞ and A . Increasing A results in a wider waverider. We consider an inviscid flow with the real gas models used previously. We first consider $M_\infty = 5$, and A is varied between 0.25 to 1.5. The second Mach number considered is $M_\infty = 6.0$, and A is swept from 0.775 to 1.0. The third Mach number considered is $M_\infty = 7.0$, and A is swept from 0.60 to 1.5. The relationship between A and volume displacement, $\forall$, is linear.

Figure 7a shows predictions of the hypersonic near-field at $rL^{-1} = 0.1667$, and the sonic boom at the ground propagated through a standard atmosphere with no wind from altitude of 15.8 km. Here, lift coefficient, c_l is calculated by integrating vehicle surface pressure over its surface area and normalizing by $(1/2)\gamma M_\infty^2 p_\infty S_v$, where S_v is the spanwise area. The principal observation from the near-field in Figs. 7a, 7b, and 7c is the positive relationship between maximum overpressure and $\forall$. With increasing volume, the strength of the leading shock wave increases. The resultant Prandtl-Meyer expansion wave also increases in strength. In the plots of near-field overpressure, in cases with successively larger $\forall$, there is a lower minimum $\Delta p p_\infty^{-1}$ proceeding the trailing shock wave.

Trends in sonic boom overpressure with increasing volume displacement are examined. These are presented in Figs. 7a, 7b, and 7c. A trend is most apparent in Fig. 7a at $M_\infty = 5$. In this case, the maximum overpressure of $\Delta p = 6.26$ Pa when $\forall = 4504.2 \text{ cm}^3$ in contrast to $\Delta p = 2.73$ Pa when $\forall = 750.7 \text{ cm}^3$. This is an approximate 129% increase in sonic boom maximum overpressure for a 507% increase in vehicle volume displacement. Figure 8a shows the relationship between the flight-vehicle volume displacement and c_l against A . Volume displacement, indicated by the solid black line, is linearly related with A . Calculations of c_l increase with increasing A . Spurious solutions for $A = 0.65$ for $M_\infty = 5$, $A = 0.85$ for $M_\infty = 6$, and $A = 0.90$ for $M_\infty = 7$ were removed to improve the visualization of c_l trends with A . Figures 8b through 8d show the extracted maximum near-field and sonic boom overpressures against the waverider's volume displacement. We show that near-field and sonic boom overpressures, via linear regression, increase with $\forall$ at $1.42 \times 10^{-1} \text{ cm}^3 \text{ kg}^{-1}$, and $8.57 \times 10^{-4} \text{ Pa cm}^3 \text{ kg}^{-1}$ for $M_\infty = 5.0$ and $2.17 \times 10^{-4} \text{ cm}^3 \text{ kg}^{-1}$, and $7.89 \times 10^{-4} \text{ Pa cm}^3 \text{ kg}^{-1}$ for $M_\infty = 7.0$.

A notable anomaly for $M_\infty = 7$ in Fig. 8d occurs when $\forall = 2402 \text{ cm}^3$ and $c_l = 0.015$. Here, there is an increase in near-field overpressure to $\Delta p p_\infty^{-1} = 1.02$, sonic boom overpressure decreases to $\Delta p = 5.18$ Pa. There is also a small decrease in sonic boom overpressure when $\forall = 2982.0 \text{ cm}^3$ to $\Delta p = 6.04$ Pa. For the cases when $M_\infty = 6$ in Fig. 8c, we do not observe the same trend. For the cases with the most extreme difference in volume displacement, near-field $\Delta p p_\infty^{-1}$ and sonic boom overpressure only vary by 0.13 and 0.47 Pa, respectively.

IV. Conclusion

A parametric study of near-field and sonic boom signatures by varying waverider width parameter, A , and M_∞ is performed. Several power-law waverider geometries of varying spanwise width at $M_\infty = 5, 6$, and 7 are studied to determine the relationship between near-field and sonic boom overpressure with volume displacement and lift coefficient. The UPS code of Lawrence et al. [17] is used to march solutions of the PNS equations downstream over hypersonic waveriders. The near-field is extracted at the under-track location and is used as the source of the propagation solvers. Near-field waveforms are propagated from constant altitudes of 15.8 km to the ground. The effect of thermo-viscous losses and absorption in the atmosphere are studied with PCBurq and the enhanced Burgers' solver within PCBoom.

The effects of waverider volume displacement and lift on near-field and sonic boom overpressures are quantified and trends are extracted. The rate of increase in near-field and sonic boom overpressure with $\forall$ is $1.42 \times 10^{-1} \text{ cm}^3 \text{ kg}^{-1}$, and $8.57 \times 10^{-4} \text{ Pa cm}^3 \text{ kg}^{-1}$ for $M_\infty = 5$ and $2.17 \times 10^{-4} \text{ cm}^3 \text{ kg}^{-1}$, and $7.89 \times 10^{-4} \text{ Pa cm}^3 \text{ kg}^{-1}$ for $M_\infty = 7$. There is a positive quasi-linear relationship between volume displacement and overpressure, in the near-field and at the ground. For example, for the minimum and maximum vehicle volumes studied at $M_\infty = 7$, there is an increase of maximum overpressure by 92.5% and 60.9% in the near-field and at the ground, respectively, when vehicle volume is increased by 150%. The SPL spectra at the ground is computed to identify frequency domain trends in the ground signatures. The power law relationship between the SPL spectra of hypersonic sonic boom, invariant to M_∞ , goes as $-19.7 \log f$.

Future work will emphasize separating volume displacement and lift in the near-field and sonic boom signature. Thermo-viscous losses and atmospheric absorption will be examined. For extreme cases with $\forall = 4970 \text{ cm}^3$ and $M_\infty = 7.0$, maximum Δp reductions using PCBurq and enhanced Burgers' solvers compared to WPM are 41.6% and 39.5%. Atmospheric loss inclusion in sonic boom models can halve maximum overpressure. Accurate hypersonic sonic boom predictions necessitate incorporating these effects.

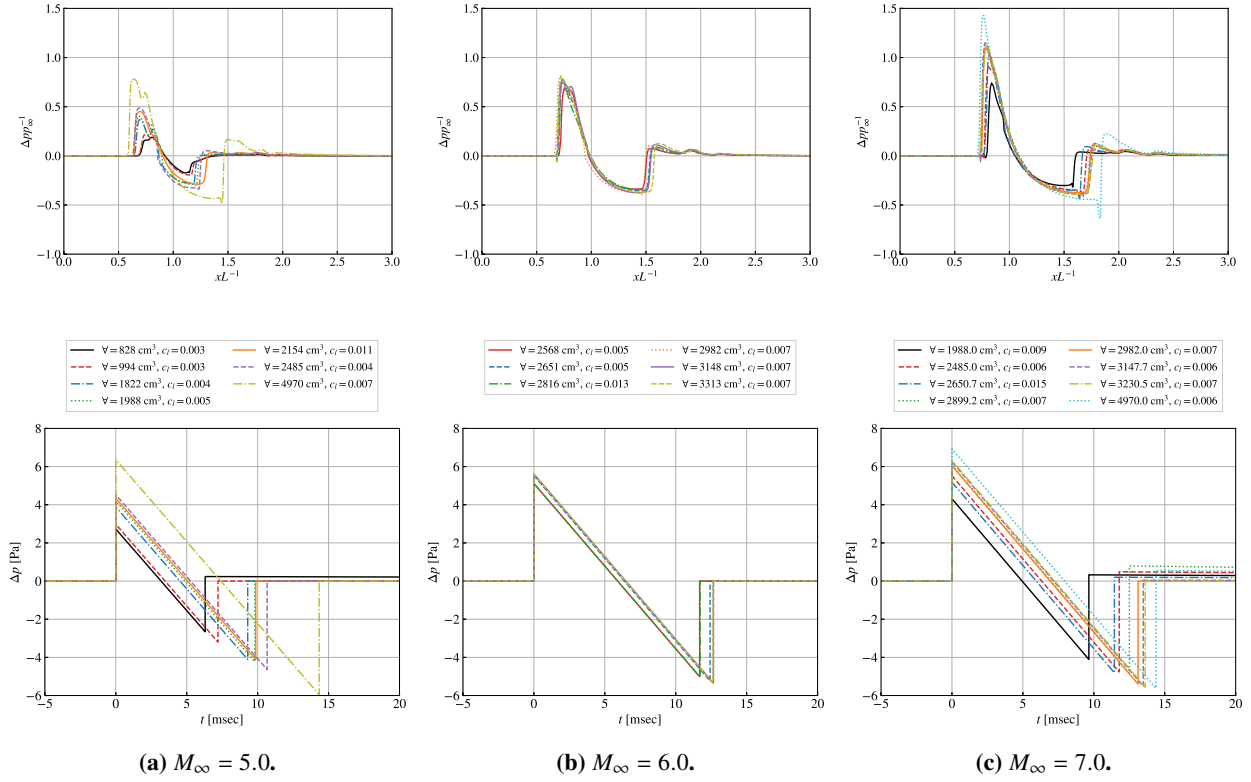


Fig. 7 Predictions of near-field (top) and sonic boom (bottom) waveforms.

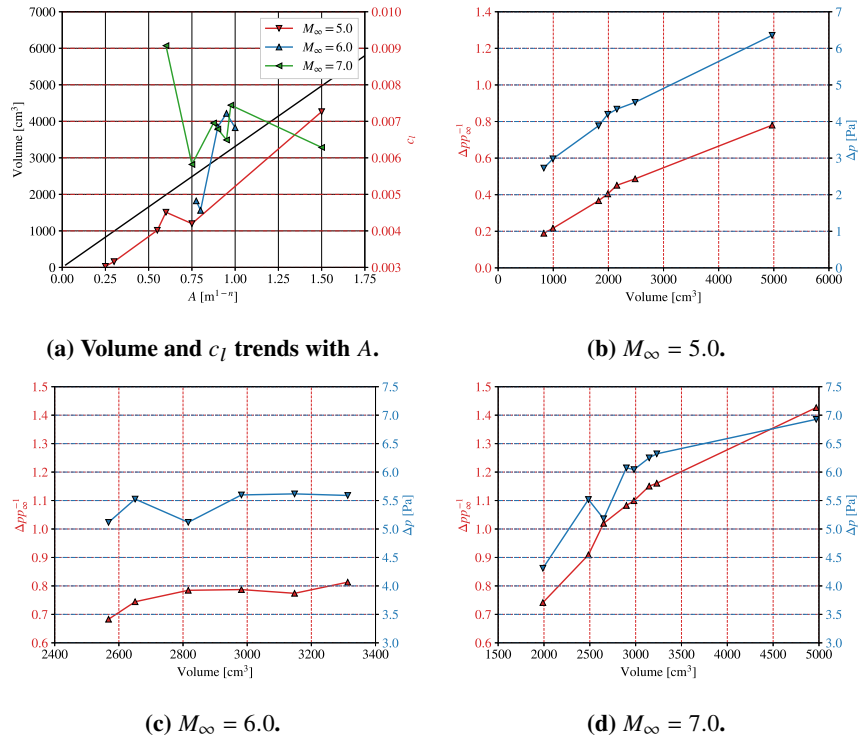


Fig. 8 Trends of near-field and sonic boom overpressures with volume displacement and lift.

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